

Governance / Constitution

Debt Status Audit

Date: 2026-06-14 **Scope:** Read-only verification of every §6.*-debt entry + CM-* paper-only/owed gate in root CLAUDE.md, cross-checked against the actual code/scripts/hooks at HEAD 357f867a. **Method:** Each row's "actual-status" is verified against a real artifact (file existence, hook line, verify-all wiring), not against the CLAUDE.md prose. Where CLAUDE.md prose disagrees with the artifact, the row is flagged STALE. **Anti-bluff (§11.4.6):** Every claim is cited to a file/line or "absent". No guessed status.

Debt-item status table

debt-id	requires	actual-status	evidence (file:line / a
§6.X-debt	Emulator PID inside podman/docker container managed by pkg/runtime; --runner=containerized for gate runs on every OS	PARTIAL	submodules/containers/p containerized.go exists kvmDevicePath/kvmAvaila (containerized.go:14-35 gate path on macOS st RESOLVED_RUNNER="host- (scripts/run-challenge- matrix.sh:198,215). Lin x86_64+KVM path wo gate-host BLOCKED. S practice by §6.AH-debt
§6.Y-debt	Pre-push hook rejects code-touching commit after a distribute that did not bump versionCode	CLOSED	.githooks/pre-push:128 implements §6.Y bump ordering against last- CLAUDE.md still lists STALE.
§6.Z-debt	firebase-distribute Phase-1 Gate 6 (evidence file: matching SHA + ≤24h + BUILD SUCCESSFUL) + pre-push check +	PARTIAL	Pre-push Check 7 enfo evidence-file presence advance (.githooks/pre push:159-193). But scri distribute.sh has NO r 6 that validates SHA/ti BUILD-SUCCESSFUL comment reference (fi

debt-id	requires	actual-status	evidence (file:line / a
	tests/firebase/ hermetic test		distribute.sh:40-41). tes firebase/ exists but tes monotonicity/pepper, r evidence gate (tests/f
§6.AA-debt	Default --debug- only; reject -- release-only without companion debug evidence; split last-version- {debug,release}; tests/firebase/ hermetic test	CLOSED	firebase-distribute.sh MODE="debug" default; :3 rejects --release-only v version-debug < current pointers :150,171. CLA lists PARTIAL → STALE
§6.AB-debt	Detekt/go-vet rule for per-feature anti- bluff completeness (rendering/state- machine/gating)	OPEN	No Detekt rule; config contains only detekt.ym detekt/ listing). The re check-challenge-discri exists+STRICT but cov Challenge falsifiability the §6.AB completeness
§6.AC-debt	Detekt rule config/ detekt/lava-non- fatal-required.kts + go-vet lava-api- go/scripts/check- non-fatal- coverage.sh + pre- push + tests/non- fatal-coverage/	OPEN	config/detekt/lava-non required.kts ABSENT; scripts/check-non-fatal ABSENT; tests/non-fatal ABSENT. A bash scanne check-non-fatal-coverage (Kotlin-only, STRICT) b + Detekt custom rule - tests are not built.
§6.AD-debt (1) per-submodule pointer propagation	## INHERITED FROM block in all per- scope CLAUDE.md/ AGENTS.md/ CONSTITUTION.md	CLOSED	scripts/check-constitu 6.AD(4) enforces 54 in (docs/helix-constitutio gates.md:56).
§6.AD-debt (2) CM-COMMIT-DOCS- EXISTS	verify referenced docs in commit messages resolve	CLOSED	scripts/check-commit-d exists + wired in verify verify-all-constitution rules.sh:190). CLAUDE OWED → STALE.
§6.AD-debt (2) CM-SCRIPT-DOCS- SYNC	script ↔ docs/ scripts/*.md sync gate	CLOSED	scripts/check-script-d exists + verify-all :187; push Check 9 (.github push:195-212). CLAUDE OWED → STALE.
§6.AD-debt (2) CM-SUBAGENT- DELEGATION-AUDIT	subagent delegation audit trail	CLOSED (scanner)	scripts/check-subagent audit.sh exists + verify

debt-id	requires	actual-status	evidence (file:line / a
§6.AD-debt (2) CM-COMMIT-DOCS- EXISTS/others equivalence- mapped	CM-ITEM-STATUS/TYPE- TRACKING, CM-FIXED- COLUMN-ALIGNMENT, operator-blocked gates	CLOSED-BY- EQUIVALENCE	Gate-doc + CLAUDE.m paper-only → STALE. §6.AD.3 Path B maps t ci-evidence/ ledgers (d constitution- gates.md:14,16,19-22,30 Intentional; no scanner
§6.AD-debt (5) build-resource stats tracker	CM-BUILD-RESOURCE- STATS-TRACKER	CLOSED	scripts/build-stats-sar scripts/build-stats-rep registry exist (docs/heli constitution-gates.md: CLAUDE.md §6.AD-de OWED → STALE.
§6.AE-debt	check-challenge- coverage.sh strict- mode flip + per- feature backfill	CLOSED (strict)	scripts/check-challeng coverage.sh:39 STRICT=" {LAVA_CHALLENGE_COVERA (STRICT default); verifi it (:163). Gate-doc says STALE. Runner host-g §6.AH, not §6.AE.
§6.AF-debt (2) chaos/stress §11.4.85 (LVA-7)	runnable chaos/ stress beyond phase-1 lava-api-go	PARTIAL	scripts/run-chaos-stres (63 lines, scaffold); doc stress/DESIGN.md + EVID phase1.md exist. NOT w verify-all (no chaos gate verify-all-constitution Phase-2+ owed.
§6.AF-debt (LVA-6) codegraph own- org submodule inclusion	§11.4.79 — index own-org submodule SOURCE	CLOSED	.codegraph/config.json records LVA-6 CLOSED (1,842 submodule files cross-submodule prob blanket submodules/** e removed. CLAUDE.md OWED → STALE.
§6.AF-debt (4) per-clause CM- COVENANT-114-* -PROPAGATION gates	literal-anchor scan gates for §11.4.79- 106	OPEN	Only CM-COVENANT-114-1 row present (docs/heli constitution-gates.md: clause propagation sca new 79-106 anchors.
§6.AF-debt (5) §11.4.80 codegraph sync automation	auto git submodule init before index	OPEN	Precondition document .codegraph/config.json automation script that submodule init before c index.
§6.AG (no separate -debt)		PARTIAL	Policy is in CLAUDE.m runner enforced via ru matrix.sh; but the no-h

debt-id	requires	actual-status	evidence (file:line / a
	Containers-driven emulators, no live ADB device		requirement is superseded by §6.AH and §6.AH-debt.
§6.AH-debt	Container/VM emulator path that BOOTS on macOS (no-KVM TCG mode + macOS /proc-reap fix in Containers submodule); no host-direct fallback	OPEN	containerized.go hard-fails if no kvm for acceleration (containerized.go:19-21). no-KVM software-emulation path in run-challenge-matrix.sh resolves macOS→host-bridge. Incident recorded .lavas/evidence/sixth-law-incidents/2026-06-03-emulator-blocked/offline.json (present). No canary + client gate hook. BLOCKED on this macOS.
§6.AI-debt (1) §11.4.128 device recorder	background Genymotion/emulator recorder w/ deterministic layout	PARTIAL/ CLOSED?	scripts/record-device.sh exists; tests/device-recorder.sh exists. Whether it implements full deterministic YYYY-MM-DD-hash>/... layout + gitignore. codegraph-exclude was not verified — marked PENDING pending deeper check.
§6.AI-debt (2) §11.4.140 LAYER 2 action-prefix hook	UserPromptSubmit hook + generated slash commands	CLOSED (hook)	scripts/hooks/action-prefix-expand.sh exists + wiremock/settings.json:14-22 (UserPromptSubmit); config/actions/registry.yaml present. Generated per-agent slash commands not verified. CLAUDE.md lists OWE for the hook portion.
§6.AI-debt (3) §11.4.141 thin-index restructure + token harness	de-dup 90K-token CLAUDE.md to index lines	OPEN	No evidence of a thin-index restructure; root CLAUDE.md is a full-body document.
§6.AI-debt (4) per-clause CM-COVENANT-114-{128..141}-PROPAGATION gates	literal-anchor gates for §11.4.128-141	OPEN	Not present in docs/header/constitution-gates.md gate inventory.
CM-COVERAGE-LEDGER	§11.4.25 — strict-flip after baseline (Phase 7-debt)	CLOSED (strict)	scripts/check-coverage.sh STRICT=\$ {LAVA_COVERAGE_LEDGER_VERIFY-all :156 invokes

debt-id	requires	actual-status	evidence (file:line / a
CM-LVA-TICKETS-SYNC → CM-WORKABLE-ITEMS-SYNC	canonical workable-items binary; strict-flip	PARTIAL	Gate-doc + CLAUDE.m "advisory / STRICT flip STALE. scripts/check-workable exists + verify-all :202. still names the retired TICKETS-SYNC/scripts/ch tickets.sh (docs/helix- gates.md:25) → STALE; §6.AF notes migration done. Upstream update + export pipeline OWE blocking).
CM-COVENANT-114-16-PROPAGATION + item-tracking paper-only gates	§11.4.15/16/19/21 trackers	CLOSED-BY-EQUIVALENCE	Intentional Path-B map helix-constitution- gates.md:14,19-24). Not

STALE-claim findings (CLAUDE.md/ gate-doc says OPEN but artifact is CLOSED)

These are documentation drift, not real work. A master plan should NOT schedule work for them — only doc corrections:

1. **§6.Y-debt** — pre-push Check 6 is live (.github/pre-push:128). CLAUDE.md lists OWED.
2. **§6.AA-debt** — default flipped to --debug-only; release-only companion check live (firebase-distribute.sh:46,199). CLAUDE.md lists PARTIAL.
3. **§6.AD-debt items 2** — check-commit-docs-exists.sh, check-script-docs-sync.sh, check-subagent-delegation-audit.sh all exist + verify-all-wired + pre-push Check 9. Gate-doc marks several paper-only.
4. **§6.AD-debt item 5** — build-stats tracker shipped (build-stats-sample.sh/build-stats-report.sh). CLAUDE.md still lists OWED.
5. **§6.AE-debt** — check-challenge-coverage.sh is STRICT-default (:39). Gate-doc says advisory.
6. **CM-COVERAGE-LEDGER** — STRICT-default (check-coverage-ledger.sh:41, verify-all --strict). CLAUDE.md + gate-doc say advisory / Phase-7-debt OWED.
7. **LVA-6 (codegraph submodules)** — CLOSED per .codegraph/config.json:3. CLAUDE.md §6.AF lists OWED.

8. **§6.AI LAYER 2 action hook** — wired in .claude/settings.json:14. CLAUDE.md lists OWED.
9. **docs/helix-constitution-gates.md itself is stale** — "Last updated 2026-05-16"; predates §6.AF/AG/AH/AI (May 31 – Jun 9). Several rows describe pre-strict-flip state.

Truly-open debts (real work to schedule)

debt-id	one-line	effort	blocking
§6.AH-debt	macOS container/VM (TCG no-KVM) emulator path that actually boots; no host-direct fallback	L	YES
§6.X-debt	container-bound emulator gate path on macOS (folds into §6.AH)	L	YES
§6.AC-debt	Detekt lava-non-fatal-required.kts + Go-vet check-non-fatal-coverage.sh + tests/non-fatal-coverage/	M	no
§6.AB-debt	Detekt/go-vet per-feature anti-bluff completeness rule	M	no
§6.Z-debt (residual)	runtime Gate 6 inside firebase-distribute.sh (SHA/24h/BUILD-SUCCESSFUL) + hermetic test	M	partial
§6.AF-debt chaos (LVA-7)	chaos/stress §11.4.85 beyond phase-1; wire into verify-all	M	no
§6.AI-debt §11.4.141	thin-index restructure of CLAUDE.md + token-reduction harness	L	no
§6.AF/AI per-clause CM-COVENANT-**-PROPAGATION gates	literal-anchor scanners for §11.4.79–141	S	no
§6.AF-debt §11.4.80	auto git submodule init before codegraph index	S	no
§6.AI-debt §11.4.128		S	no

debt-id	one-line	effort	blocking
	verify/complete deterministic device- recorder layout		

Prioritized recommendations

1. **§6.AH-debt + §6.X-debt (TOP — both BLOCKING)**. Single coordinated effort in the Containers submodule: add a no-KVM TCG software-emulation runner + the macOS /proc-reap fix so the container/VM emulator path boots on darwin/arm64. Until this lands, EVERY §6.AE matrix gate, §6.Z release canary, and tag-time attestation on the operator's macOS host is honestly BLOCKED — this is the single highest-leverage debt because it unblocks the entire release pipeline.
2. **§6.Z-debt residual (firebase Gate 6)**. Add the runtime evidence gate inside firebase-distribute.sh (the pre-push Check 7 only fires on the commit that advances the pointer, not at distribute time). Cheaper than #1 and closes a real distribute-faulty-version vector.
3. **§6.AC-debt (non-fatal telemetry lint)**. Build the Detekt custom rule + Go-vet check + tests/non-fatal-coverage/. Currently only a Kotlin bash scanner exists; the Go side has zero enforcement. Pairs naturally with §6.AB-debt (same Detekt-rule-authoring effort).

Doc-hygiene action (not scheduled work, but should precede the plan): correct the 9 STALE claims above in CLAUDE.md + regenerate docs/helix-constitution-gates.md (stale since 2026-05-16) so the master plan is built on accurate status. Several debts are already done.

Open vs closed count: Of ~27 distinct debt/gate items examined: **10 truly-open** (2 BLOCKING, 1 partial-blocking, 7 non-blocking), **3 PARTIAL**, ****~14 CLOSED or closed-by-equivalence**** (9 of which are STALE-marked-open in CLAUDE.md/gate-doc).